Worksheet — Circular Motion Simulation

Use with *Simulation - Circular motion.html* · circular motion (Block 5): rotating chair, satellite, car on a bend, playground carousel, gears, loop-the-loop, vertical circle, conical pendulum

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/16-circular-motion** in a browser (offline: *Simulation - Circular motion.html*). Choose an example at the top: 🪑 rotating chair · 🛰️ satellite · 🚗 car on a bend · 🎠 carousel. Set values with ▼▲, or type a value in the box.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min). The Tasks tab has the same three parts.

Use g = 9.81 N kg^{-1}, G = 6.67 × 10^{-11} N m^2 kg^{-2}, Earth: M = 5.97 × 10^{24} kg, R = 6.37 × 10^6 m.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🚗 A car goes round a bend at constant speed. Which way does its acceleration point?			
2. Which real force pushes the car towards the centre on a flat road?			
3. Same bend, twice the speed. The friction needed is × ____ ?			
4. 🎠 Child C lets go of the handle. Which path does C take?			
5. 🪑 Same ω: arms open (0.75 m) instead of closed (0.20 m). The arm's pull is × ____ ?			
6. 🛰️ A satellite moves to a higher circular orbit. Faster or slower?			

7. 🚗 Draw the free-body diagram of a 1200 kg car on a **flat** bend of radius 50 m at 15 m s^{-1}, seen from behind (the centre of the bend is on the left). Then calculate the friction force it needs. Check with the FBD in the simulation.

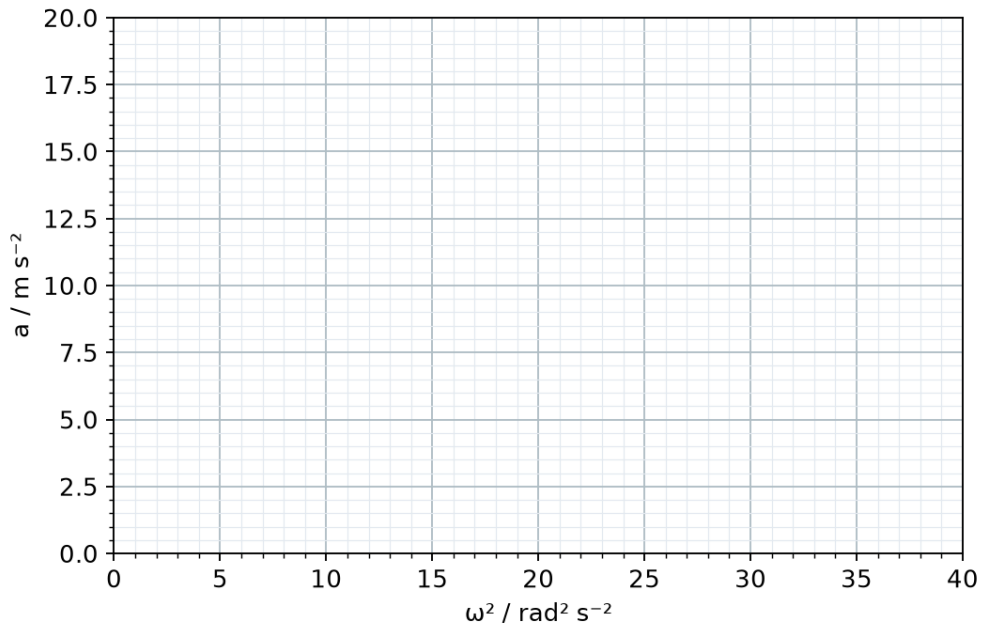
Free-body diagram and working

Part 2 — During the lab

A · Phone on a spinning chair (phyphox). Simulation: 🪑 chair, Motor mode, mass in each hand m = 0.20 kg (the phone), arms open r = 0.50 m. Type each period T in the box, then read the acceleration of the hand (arms open).

T / s	$\omega = 2\pi/T$ / rad s^{-1}	ω^2 / rad^2 s^{-2}	simulation a / m s^{-2}	class phone a / m s^{-2}
1.0				
1.5				
2.0				
2.5				
3.0				

8. Plot a against ω^2 and draw the best-fit line. Gradient = _____ . What does the gradient represent?



9. Compare your gradient with the distance of the class phone from the axis of the chair. Give one reason why the phone's graph is not a perfect straight line.

B · Playground carousel. Simulation: 🎠 carousel, T = 4.0 s, three children of 30 kg at 0.60 m, 1.20 m and 1.80 m. Strongest grip 150 N.

child	r / m	$v = \omega r$ / m s^{-1}	grip needed $F = m\omega^2 r$ / N	holds on?
A	0.60			
B	1.20			
C	1.80			

10. Calculate the shortest period T at which child C can still hold on (grip 150 N). Then lower T in the simulation in steps of 0.1 s and find when C lets go. Describe C's path after letting go.

Working

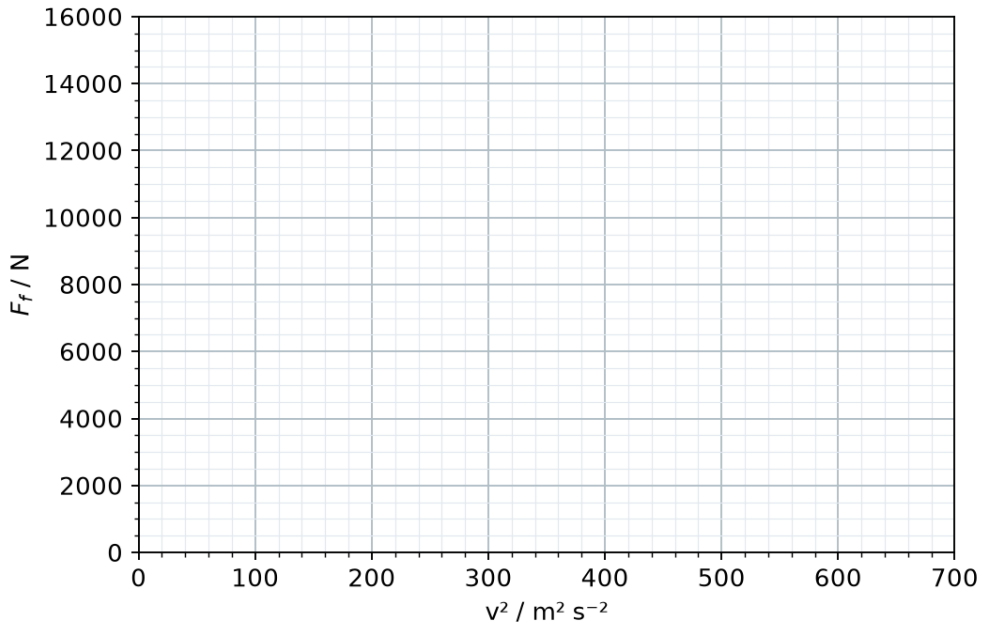
C · Extension (beyond SL): the rotating-chair demo. Simulation: Free spin ★, T = 3.0 s with arms open, 1.0 kg in each hand. Press ▶️ Start, then 🤜 Pull arms in.

11. Read ω and T before and after pulling the arms in, and the kinetic energy. Explain why ω increases, and where the extra kinetic energy comes from.

Part 3 — Homework

12. 🚗 Flat road, 1200 kg car, r = 50 m, dry asphalt. Record the friction needed at each speed, then plot F_f against v^2.

v / m s^{-1}	5	10	15	20	25
v^2 / m^2 s^{-2}					
F_f / N					



Gradient = _____ . Show that the gradient equals m/r.

13. On wet asphalt (μ_s = 0.50), calculate the maximum speed on the 50 m bend. Would a 2400 kg van have a different maximum speed? Explain.

Working

14. The ISS orbits 410 km above the Earth's surface. Calculate its orbital speed and its period (in minutes).

Working

15. A geostationary satellite has a period of 24 h. Calculate the radius of its orbit and its height above the surface.

Working

16. Choose each orbit from the Orbit list and read the period. Complete the table and plot T^2 against r^3 (or use the Data tab). Use the gradient to find the mass of the Earth.

orbit	h / km	r / m	T / s	r^3 / m^3	T^2 / s^2
ISS	410				
Hubble	540				
GPS	20 200				
Geostationary	35 857				

Gradient and mass of the Earth

17. 🪑 Motor-driven chair, T = 1.5 s, arms open (0.75 m), 1.0 kg in each hand. Draw the FBD of one mass and calculate the inward force the arm exerts on it, and the total force of the hand.

Free-body diagram and working

18. Your friend says: "On the carousel a centrifugal force throws you outwards." Use the free-body diagrams in the simulation to explain what really happens.

Part 4 — Gears: v = ωr passed from wheel to wheel

Simulation: ⚙️ Gears. Kinematics only (no forces). Every gear has teeth of the same size (module 4 mm), so its radius is $r = N \times 2$ mm. A coloured tooth on each gear lets you count turns; the readouts give rpm, ω, T and the rim speed $v = \omega r$.

A · Predict (before you press ▶️ Start).

Question	My prediction (before)	What the simulation showed	Physics explanation
19. Gear A (12 teeth) at 60 rpm drives gear B (36 teeth). How fast does B turn, and which way?			
20. Which is the **same** for A and B: the angular speed ω, the period T or the rim speed v?			

B · During the lab. Set-up: Gear train. A (12 teeth) at 60 rpm → B (24 teeth, idler) → C (36 teeth); D (12 teeth) is fixed on C's shaft.

21. Read the table under the scene and complete this table.

gear	*teeth N*	*r / cm*	*rpm*	*ω / rad s^{-1}*	*v = ωr / cm s^{-1}*
A (driver)					
B (idler)					
C (output)					
D (on C's shaft)					

22. Use your table: which quantity is the same for A, B and C? Which is the same for C and D? Explain both with $v = \omega r$.

23. Change the idler B from 24 to 48 teeth. What happens to the speed and the direction of C? What does an idler gear do?

C · Homework.

24. 🚲 Bicycle, pedalling at 80 rpm, wheel radius 0.34 m. Use the simulation (Bicycle) or calculate $v = \omega_{pedals} \times (N_{chainring}/N_{sprocket}) \times 0.34$ m.

gear (chainring/sprocket)	44/16	50/12	34/28	39/21	44/11
gear ratio N_c/N_s					
bike speed / m s^{-1}					

Which gear would you choose to climb a steep hill, and which on a fast flat road? Explain using $v = \omega r$.

25. Motor + belt: the motor turns at 300 rpm with a 2.0 cm pulley; the belt drives a 10 cm pulley, and a 3.0 cm drum on the same shaft winds up a rope. Calculate the speed at which the load rises and the time to lift it by 50 cm. Check with the ruler in the simulation.

Working

Part 5 — Vertical circles and the conical pendulum

Simulation: ↕ Vertical → 🎢 Loop-the-loop; ↗ At an angle → ⛓ Conical pendulum. In a vertical circle energy is conserved: the object is slowest at the top and fastest at the bottom. The FBD shows the forces *now* and, under it, the forces at the **top · side · bottom** to the same scale. The conical pendulum turns in a horizontal circle; the tension is written F_T because T is the period.

A · Predict (before you press ▶️ Start).

Question	My prediction (before)	What the simulation showed	Physics explanation
26. 🎢 A rider goes round a vertical loop. Where does the seat push hardest on the rider: top, side or bottom?			
27. ⛓ The swing ride turns faster (shorter period). Do the chains hang more steeply or swing out further?			

B · During the lab. 🎢 Loop-the-loop (↕ Vertical): 60 kg, r = 8.0 m, v_{top} = 12 m s^{-1} (the start values).

28. Read the three snapshots under the FBD (and the readouts) and complete the table.

position	v / m s^{-1}	F_N / N	F_N / mg	equation (towards the centre +)
top				
side				
bottom				

29. Calculate F_N at the top, and the slowest speed at the top for which the rider stays on the track. Then set v_{top} = 7 m s^{-1}, press ▶️ Start and describe what happens.

Working

C · Homework.

30. ⛓️ Conical pendulum (↗ At an angle): 2.0 kg seat, L = 1.5 m. Record trials with the period T = 1.5, 1.8, 2.0 and 2.29 s. Complete the table, then plot cos θ against T^2 (Data tab). Use the gradient to find g.

T / s	T^2 / s^2	θ / °	cos θ	F_T / N
1.5				
1.8				
2.0				
2.29				

Gradient and g

31. Draw the free-body diagram of the seat. Split F_T into parts and show that $F_T = m\omega^2 L$ and $\cos\theta = g/(\omega^2 L)$.

Free-body diagram and working

32. Explain why the chains can never be horizontal, however fast the ride turns.